

Determinants

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Determinants assign a signed volume and test invertibility. Their defining properties yield row-operation rules, cofactor expansions, permutation formulas, and inverse formulas.

DEFINITION AND PROPERTIES

$$\det I = 1, \quad \det(AB) = \det(A) \det(B), \quad A \text{ invertible} \iff \det A \neq 0.$$

Row rules:

swapping rows changes the sign; multiplying a row by c multiplies the determinant by c ; adding a multiple of one row to another leaves it unchanged.

Linearity:

the determinant is linear in each row (or column) separately and alternating in its rows.

$$\det(A^T) = \det A, \quad \det(A^{-1}) = \det A^{-1}, \quad \det(cA) = c^n \det A.$$

COMPUTATIONAL FORMULAE

Triangular matrices:

$$\det A = \prod_{i=1}^n a_{ii}.$$

$$\det A = \sum_{j=1}^n a_{ij} C_{ij} = \sum_{i=1}^n a_{ij} C_{ij}, \quad C_{ij} = (-1)^{i+j} \det(A_{ij}).$$

Cofactor formula:

$$A^{-1} = \frac{1}{\det A} \text{Cof}(A)^T.$$

$$\det A = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{i=1}^n a_{i, \sigma(i)}.$$

Cramer's rule:

$$\text{if } Ax = b \text{ and } \det A \neq 0, \quad x_i = \frac{\det A_i(b)}{\det A}.$$

Vandermonde determinant:

$$\det \left([x_j^{i-1}]_{i,j=1}^n \right) = \prod_{1 \leq i < j \leq n} (x_j - x_i).$$

GEOMETRY AND VOLUME

$$\text{volume}(A([0, 1]^n)) = |\det A|, \quad \det A = 0 \iff \text{columns lie in a lower-dimensional subspace.}$$

PROBLEM SET

Q1.

Use row operations to compute

$$\det \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ 1 & 1 & 1 \end{bmatrix}.$$

State the determinant rule used at each step and identify the product of the pivots.

Q2.

Let A and B be 4×4 matrices with $\det A = -2$ and $\det B = 3$.

- (i) Find $\det(AB)$ and $\det(A^T)$.
- (ii) Find $\det(A^{-1})$, $\det(A^2)$, and $\det(2A)$.

Q3.

Let

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 2 \\ 0 & 4 & 1 \end{bmatrix}.$$

- (i) Find the cofactor matrix $C = (C_{ij})$ of A .
- (ii) Use a cofactor expansion to find $\det A$.
- (iii) Verify the identity $AC^T = (\det A)I$.

Q4.

For

$$A(t) = \begin{bmatrix} 1 & 1 & 1 \\ 1 & t & 2 \\ 1 & 2 & t \end{bmatrix},$$

- (i) Compute $\det A(t)$ by row operations.
- (ii) For which values of t is $A(t)$ invertible?
- (iii) When $t = 0$ or $t = 2$, give a nonzero vector in the nullspace of $A(t)$.

Q5.

Use Cramer's rule to solve

$$2x + y = 1, \quad x + 2y + z = 2, \quad y + 2z = 3.$$

Compute $\det A$ and the three determinants obtained by replacing one column of A by b .

Q6.

Use the cofactor formula to find the inverse of

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$$

Verify the result by multiplication.

Q7.

- (i) Find the area of the parallelogram spanned by $u = (3, 2)$ and $v = (1, 4)$.
- (ii) Find the volume of the parallelepiped spanned by

$$u = (1, 1, 0), \quad v = (1, 0, 1), \quad w = (0, 1, 1).$$

Q8.

Suppose that A is an invertible matrix with integer entries.

- (i) Prove that if $\det A = 1$ or $\det A = -1$, then every entry of A^{-1} is an integer.
- (ii) Prove the converse: if both A and A^{-1} have integer entries, then $\det A = 1$ or $\det A = -1$.